Hamiltonian Graphs and Digraphs

Theorem 3 (Ore, 1960)

If G is a simple graph with n (≥ 3) vertices, and $\deg(v) + \deg(w) \geq n$ for every pair of non-adjacent vertices v and w, then G is Hamiltonian.

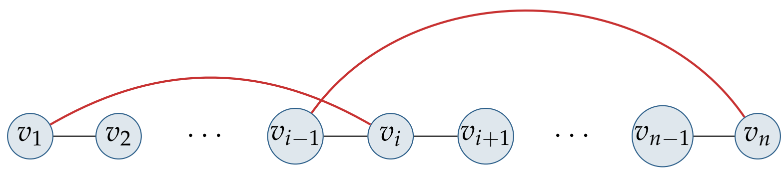
Proof. We assume the statement is false and derive a contradiction. So let G be a non-Hamiltonian graph on n vertices satisfying the given condition on the vertex degrees. By adding extra edges if necessary, we may assume that G is "just barely" non-Hamiltonian, in the sense that adding any further edge produces a Hamiltonian graph (note that adding extra edges cannot violate the degree condition, since degrees can only increase). It follows that G contains a path

$$v_1 \to v_2 \to \cdots \to v_n$$

passing through every vertex. But since G is non-Hamiltonian, the vertices v_1 and v_n are not adjacent, and so $\deg(v_1) + \deg(v_n) \geq n$; by the pigeonhole principle, there must be some vertex v_i adjacent to v_1 with the property that v_{i-1} is adjacent to v_n. But this gives the required contradiction, since

$$v_1 \to v_2 \to \cdots \to v_{i-1} \to v_n \to v_{n-1} \to \cdots \to v_{i+1} \to v_i \to v_1$$

is then a Hamiltonian cycle.



□

Corollary 7.2 (Dirac, 1952)

If G is a simple graph with n (≥ 3) vertices, and $\deg(v) \geq \dfrac{n}{2}$ for each vertex v, then G is Hamiltonian.

Proof. The result follows from the preceding theorem, since for all $v, u \in V(G)$ we have $\deg(v) + \deg(u) \geq n$, and hence G is Hamiltonian. □

Theorem (Hamiltonian Digraphs — Ghouila-Houri)

Let D be a strongly connected digraph with n vertices. If $\operatorname{outdeg}(v) \geq \dfrac{n}{2}$ and $\operatorname{indeg}(v) \geq \dfrac{n}{2}$ for each vertex v, then D is Hamiltonian.